Differential Equations — Reference

how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

Do	How
1. find the pieces	Each lesson opens with a PIECES table. It says where each piece is in your equation, and its color.
2. fill the slots	A piece goes in the slot of the same color: p goes in $\boxed{p}$.
3. one box at a time	Each box is one move. Do not merge two boxes.

Colors in lessons 1, 3, 5 and Bernoulli: $p(x)$ $q(x)$ μ (mu) x_0, y_0 $a(x)$ or n. Lessons 2, 4 and the homogeneous method set their own colors in their PIECES tables.

the master idea

Rearrange until each side is integrable.

Method	Rearrange it until
linear (L3)	the left side is one derivative, $(\mu y)'$
separable (L4)	all y is with dy, all x is with dx
Bernoulli (L6)	it is linear in $v = y^{1-n}$. Then L3
homogeneous (L6)	it is separable in $v = y/x$. Then L4

which method? — the sorter

Test in this order. Stop at the first yes.

If the equation	Use	
fits $y' + p(x)\,y = q(x)$ after you divide	linear, seven lines	(L3)
factors as $y' = g(x)\,h(y)$	separable	(L4)
fits $y' + p\,y = q\,y^n$, $n \neq 0, 1$	Bernoulli, $v = y^{1-n}$	(L6)
has a right side in $\frac{y}{x}$ only	homogeneous, $v = \frac{y}{x}$	(L6)

Trap: Some equations fit two rows. $y' + 3y = 6$ is linear and separable. Either method works.
Trap: Bernoulli with $n = 0$ or $n = 1$ is already linear. Do not substitute.

two meanings of "homogeneous"

Where	Means
Lesson 1 (Trench 2.1)	the right side of $y' + py = q$ is the zero function: $q = 0$
Lesson 6	the right side is a function of $\frac{y}{x}$ only: $y' = F(y/x)$

which tool? — given → find

Given → find	Do this	
an equation → order	highest derivative	(L1)
an equation → linear or not	the linear checklist	(L1)
an equation → p and q	standard form	(L1)
a y and a DE → solution or not	differentiate y, substitute	(L1)
$y' = f$ and a point → slope	evaluate f there	(L2)
f → zero-slope set	factor $f = 0$	(L2)
f with no x, y_0 → limit	nearest flat line in the drift	(L2)
linear → general solution	seven lines	(L3)
separable → general solution	separate, integrate, free y	(L4)
general solution, IC → C	plug in x_0, y_0	(L3, L4)
separable → lost solutions	roots of $h(y) = 0$	(L4)
linear IVP → interval	bad x of p and q	(L5)
nonlinear IVP → unique or not	Test 1, Test 2	(L5)
Bernoulli → solution	$v = y^{1-n}$, then L3	(L6)
$F(y/x)$ → solution	$v = y/x$, then L4	(L6)

which way? — differentiate or integrate

Task	Direction
verify a solution	differentiate the given y
check μ	differentiate μ, compare with $p\mu$
uniqueness test	differentiate f in y (x is a constant)
linear line 6, separable step 3	integrate

Trap: To verify, never solve the DE for y'. Get y' from the given y.

integrals (add $+C$ once)

Function	Antiderivative
x^n, $n \neq -1$	$\frac{x^{n+1}}{n+1}$
y^{-n}, $n \neq 1$	$\frac{y^{1-n}}{1-n}$ *(only $n = 1$ gives ln)*
$\frac{k}{x}$	$k \ln\lvert x\rvert$ not $\ln \frac{k}{x}$
$\frac{1}{ax+b}$	$\frac{1}{a} \ln\lvert ax+b\rvert$
e^{kx}	$\frac{1}{k} e^{kx}$ divide by k
$\sin(ax)$ / $\cos(ax)$	$-\frac{\cos(ax)}{a}$ / $\frac{\sin(ax)}{a}$
$\tan x$	$\ln\lvert\sec x\rvert$
$\sec^2 x$	$\tan x$
$\frac{1}{1+y^2}$	$\tan^{-1} y$

By parts: $\int u\,dv = uv - \int v\,du$. Pick u = the part that gets simpler when you differentiate (x, x^2, $\ln x$).
Product rule (the heart of L3): $(\mu y)' = \mu' y + \mu y'$.

exponent and log rules

Rule	Use
$e^{k \ln u} = u^k$	simplify μ: $e^{2\ln x} = x^2$, $e^{-\ln x} = \frac{1}{x}$
$\frac{1}{e^{-kx}} = e^{kx}$	line 7: divide by μ
$\frac{1}{\sec x} = \cos x$	line 7 when $\mu = \sec x$
$e^{a+C} = e^C e^a$	L4: rename $e^C \to C$
e^{ab}	does not split: $e^{xy} \neq e^x e^y$
$\ln(e^{\square}) = \square$	ln and e undo each other

Lesson 1 — classify, standard form, verify

EXPLANATION

THE IDEA
Label the equation before you solve it: order, linear or not, homogeneous or not. Then put it in standard form, so that you can read p and q.

Pieces

Piece	Where to find it
$a(x)$	the coefficient of y', with all y-terms on the left
$p(x)$	the coefficient of y, after you divide by a. It keeps its sign
$q(x)$	the right side, after you divide by a

Read the pieces

Quick example	Work → answer
$xy'+2y=x^2$	$a=x$: $y'+\frac{2}{x}y=x$
$y'=3y+x$	move: $y'-3y=x$. $p=-3$
$y'+y\sin x=0$	$p=\sin x$, $q=0$: homogeneous
$y'=xy^2$	y^2: not linear. No p, no q

Linear or not?

Still linear	Breaks linearity
x^2y', e^xy, $\sin x$	y^2, yy', $(y')^3$
a missing term (coefficient 0)	$\sin y$, e^y

Trap: p includes its sign. $y'-3y$ has $p=-3$.
Trap: Divide every term by a, the right side too.
Trap: Homogeneous means q is the zero function. Do not plug in $x=0$.
Trap: To verify, never solve the DE for y'. Get y' from the given y.

WORK

Steps — classify

1. ORDER
The highest derivative present. Exponents do not matter: $(y'')^3$ is order 2.

↓

2. LINEAR?
Every $y, y', y'', \ldots$ appears alone, to the first power, with a coefficient in x only. Never x-stuff breaks it.

↓

3. HOMOGENEOUS? (2.1 SENSE)
Move all y-terms to the left. If the right side is the zero function, it is homogeneous.

Steps — standard form

1. MOVE
All y-terms to the left: $\boxed{a}y'+by=g$

↓

2. DIVIDE EVERY TERM BY a
$y'+\boxed{\frac{b}{a}}y=\boxed{\frac{g}{a}}$ *(the coefficient of y' must be 1)*

↓

3. READ
$p=\frac{b}{a}$ $\quad q=\frac{g}{a}$ *(write both down)*

Steps — verify a solution

1. DIFFERENTIATE
Find y' (and y'' if needed) from the given y, with calculus.

↓

2. SUBSTITUTE
Put y, y', … into the left side only.

↓

3. SIMPLIFY AND COMPARE
Simplify the left side. If it equals the right side for every x, y is a solution.

Lesson 2 — direction fields

EXPLANATION

THE IDEA
$y' = f(x,y)$ gives the slope at every point. The field is a picture of those slopes. You never solve anything.

Pieces

Piece	Where to find it
$f(x,y)$	the right side, after you get y' alone
(x_0,y_0)	the point you are asked about

Sign of the slope

Slope	The solution
$y' > 0$	rises as x increases
$y' < 0$	falls
$y' = 0$	is flat

Does f have x?

If f	Then
has no x: $y' = f(y)$	every column is the same. Zero-slope lines are horizontal (equilibria). Solutions approach them and never cross
has x	the zero-slope set is a curve, not a wall. No equilibria

Trap: Never divide $f = 0$ by y. You lose the root $y = 0$.
Trap: The limit only works when f has no x.

WORK

Steps — slope at a point

1. TEMPLATE
slope $= \boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$

↓

2. READ THE SIGN
$+$ rises, $-$ falls, 0 flat.

Steps — zero-slope set

1. SET
$\boxed{f(x,y)} = 0$

↓

2. FACTOR
Pull out every common factor. Do not divide.

↓

3. EACH FACTOR $= 0$
Each factor gives one line or curve of the set.

Steps — limit as $x \to \infty$ (f has no x)

1. FLAT LINES
Solve $\boxed{f(y)} = 0$. List the roots.

↓

2. DRIFT
Sign of $\boxed{f}\left(\boxed{y_0}\right)$: $+$ moves up, $-$ moves down, 0 stays.

↓

3. LIMIT
The nearest flat line in the direction of drift. If there is none in that direction: $\pm\infty$.

Lesson 3 — first order linear (seven lines)

EXPLANATION

THE IDEA

Multiply by μ so that the left side becomes one product-rule derivative, $(\mu y)'$. Then one integral finishes the job.

Pieces

Piece	Where to find it
p	the coefficient of y in standard form, sign included
q	the right side in standard form
μ (mu)	you make it: $e^{\int p\,dx}$. No $+C$
x_0, y_0	the initial condition. Use it only at the end

Line 2 — common μ

If p is	then μ is
k (a constant)	e^{kx}
$\frac{k}{x}$	$e^{k\ln x} = x^k$
$-\frac{1}{x}$	$e^{-\ln x} = \frac{1}{x}$
$2x$	e^{x^2}
$\tan x$	$e^{\ln\lvert\sec x\rvert} = \sec x$
$-\tan x$	$e^{\ln\lvert\cos x\rvert} = \cos x$

Equivalences

This	is the same as
μ'	$p\mu$, always (that is line 3)
$\mu p\,y$	a line 4 term
μy	the line 6 left side
$\frac{1}{e^{-kx}}$ / $\frac{1}{\sec x}$	e^{kx} / $\cos x$

Trap: $\int \frac{k}{x}\,dx = k\ln x$, not $\ln\frac{k}{x}$.
Trap: $\int e^{kx}dx = \frac{1}{k}e^{kx}$. Divide by k.
Trap: $\int \tan x\,dx = \ln\lvert\sec x\rvert$.
Trap: Line 7: divide C by μ too.

WORK

The seven lines

LINE 1 — STANDARD FORM, READ THE PIECES

$y' + \boxed{p}y = \boxed{q}$ Write $p = \ldots$ and $q = \ldots$

↓

LINE 2 — BUILD μ

$\boxed{\mu} = e^{\int \boxed{p}dx}$ Simplify. $(e^{k\ln u} = u^k)$ No $+C$.

↓

LINE 3 — CHECK μ

Differentiate: $\boxed{\mu}' = \ldots$ Separately: $\boxed{p}\boxed{\mu} = \ldots$
They must be equal. If not, fix line 2.

↓

LINE 4 — MULTIPLY EVERY TERM

$\boxed{\mu}y' + \boxed{\mu}\boxed{p}y = \boxed{\mu}\boxed{q}$
(three multiplications, written out)

↓

LINE 5 — COLLAPSE THE LEFT SIDE

$\left(\boxed{\mu}y\right)' = \boxed{\mu}\boxed{q}$
$((\mu y)' = \mu' y + \mu y' = \mu p\,y + \mu y'$: the whole left side of line 4)

↓

LINE 6 — INTEGRATE BOTH SIDES

$\boxed{\mu}y = \int \boxed{\mu}\boxed{q}dx + C$
(left side is μ times y: not μ', not μp)

↓

LINE 7 — DIVIDE BY μ

$y = \frac{1}{\boxed{\mu}}\left(\int \boxed{\mu}\boxed{q}dx + C\right)$
(divide the whole parenthesis: C becomes C/μ)

↓

IC (IF GIVEN)

Plug in x_0 and y_0. Solve for C. Rewrite the answer with the number.

Lesson 4 — separable

EXPLANATION

THE IDEA

Factor the right side into an x-part times a y-part. Move all y to the dy side and all x to the dx side. Integrate each side.

Pieces

Piece	Where to find it
$g(x)$	the factor with only x (a constant counts)
$h(y)$	the factor with only y
x_0, y_0	the initial condition

Read the pieces

Quick example	Work → answer
$y' = xy$	$g = x$, $h = y$
$y' = \frac{x}{y}$	$g = x$, $h = \frac{1}{y}$
$y' = e^{x+y}$	$= e^x e^y$: $g = e^x$, $h = e^y$
$y' = 2y$	$g = 2$, $h = y$
$y' = x + y$	will not factor: not separable

Step 4 — free y

Wrapped	Undo with	Watch for
$\ln\lvert y\rvert = \ldots$	$e^{(\cdot)}$ of the whole side	$e^{a+C} = e^C e^a$. Rename $e^C \to C$
$y^2 = \ldots$	square root	$y = \pm\sqrt{\ldots}$. Keep $\pm$ unless an IC picks one
$\frac{1}{y} = \ldots$	reciprocal	do it first, before anything else
$\tan^{-1} y = \ldots$	tan	$y = \tan(\ldots + C)$, C stays inside

Trap: $\int y^{-n}\,dy = \frac{y^{1-n}}{1-n}$ for $n \neq 1$. Only $n = 1$ gives ln.
Trap: Keep the dy and the dx when you separate.

WORK

Steps

1. FACTOR

$\frac{dy}{dx} = \boxed{g(x)}\,\boxed{h(y)}$ If it will not factor, it is not separable.

↓

2. SEPARATE

$\frac{dy}{\boxed{h(y)}} = \boxed{g(x)}\,dx$

↓

3. INTEGRATE

$\int \frac{dy}{\boxed{h(y)}} = \int \boxed{g(x)}\,dx + C$ *(one $+C$ only)*

↓

4. FREE y

Apply the inverse of whatever wraps y. Use the table on the left.

↓

5. IC (IF GIVEN)

Plug in x_0, y_0. Solve for C.

↓

6. LOST SOLUTIONS

Solve $\boxed{h(y)} = 0$. Each root $y = k$ is a constant solution. Step 2 divided by it. Mention them if asked for all solutions.

Lesson 5 — existence and uniqueness

THE IDEA
Before you solve, ask: does a solution through the point exist, and is it the only one? Linear and nonlinear equations use different tests.

Linear: the interval

EXPLANATION

Pieces

Piece	Where to find it
p, q	standard form $y' + py = q$
x_0	the IC $y(x_0) = y_0$. y_0 does not matter

Bad x

Undefined	Bad x
division by 0	where the bottom $= 0$
$\ln(\dots)$	where $\dots \le 0$
even root $\sqrt{\dots}$	where $\dots < 0$
$\tan x$	$x = \frac{\pi}{2} + n\pi$

Read the pieces

Quick example	Work → answer
$p = \frac{1}{x-3}$, $x_0 = 1$	bad: 3. $(-\infty, 3)$
$p = \ln x$, $x_0 = 2$	bad: $x \le 0$. $(0, \infty)$

Trap: Divide by $a(x)$ first. Its zeros become bad x.

WORK

1. STANDARD FORM
$y' + \boxed{p}\,y = \boxed{q}$

↓

2. BAD x OF p
List every x where p is undefined.

↓

3. BAD x OF q
List every x where q is undefined.

↓

4. NUMBER LINE
Mark all bad x and x_0.

↓

5. INTERVAL
(nearest bad x below x_0, nearest bad x above x_0). Open. No bad x on a side: $\pm\infty$.

Nonlinear: Test 1 and Test 2

EXPLANATION

Pieces

Piece	Where to find it
$f(x,y)$	the right side of $y' = f(x,y)$
$\frac{\partial f}{\partial y}$	you make it: differentiate f in y, x held constant
(x_0, y_0)	the IC. Both numbers matter

Verdict

Result	Verdict
both pass	a unique solution
Test 1 yes, Test 2 no	exists, uniqueness not guaranteed
Test 1 fails	the theorem says nothing

Read the pieces

Quick example	Work → answer
$f = y^{1/3}$, $y_0 = 0$	$\frac{1}{3y^{2/3}}$ breaks at 0: exists, not guaranteed unique
$f = y^{1/3}$, $y_0 = 1$	both pass: unique

Trap: The tests are at the point. Changing y_0 can change the verdict.
Trap: A point on the edge of the domain fails Test 1: $\sqrt{y}$ at $y_0 = 0$ is not defined below.

WORK

1. NORMAL FORM
$y' = \boxed{f(x,y)}$

↓

2. TEST 1 — EXISTENCE
$\boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number, and f is defined on an open rectangle around the point.

↓

3. PARTIAL
$\boxed{\frac{\partial f}{\partial y}} = \dots$ *(x is a constant)*

↓

4. TEST 2 — UNIQUENESS
$\boxed{\frac{\partial f}{\partial y}}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number.

↓

5. VERDICT
Use the verdict table on the left.

Lesson 6 — transformations

THE IDEA
Substitute a new variable so that the equation becomes a type you can already solve. Bernoulli → linear (L3). Homogeneous → separable (L4).

Bernoulli: $y' + p\,y = q\,y^n$

EXPLANATION

Pieces

Piece	Where to find it
p	the coefficient of y on the left
q	the coefficient of y^n on the right
n	the power on the right. $n \neq 0, 1$
new p, new q	$(1-n)\,p$ and $(1-n)\,q$ (step 5)

Read the pieces

Quick example	Work → answer
$y' + y = xy^3$	$1, x, 3$: $v = y^{-2}$. new -2, new $-2x$
$y' - \frac{y}{x} = y^2$	$-\frac{1}{x}, 1, 2$: $v = y^{-1}$. new $\frac{1}{x}$, new -1

Trap: Keep the $(1-n)$. It is often negative.
Trap: Divide every term by y^n, the y' term too.
Trap: Forget the $(1-n)$ in step 5 and every later line is wrong.

WORK

1. MATCH, READ THE PIECES
$y' + \boxed{p}\,y = \boxed{q}\,y^{\boxed{n}}$

↓

2. CHOOSE v
$v = y^{1-\boxed{n}} \qquad v' = (1-\boxed{n})\,y^{-\boxed{n}}\,y'$

↓

3. DIVIDE EVERY TERM BY y^n
$y^{-\boxed{n}}y' + \boxed{p}\,y^{1-\boxed{n}} = \boxed{q}$

↓

4. REPLACE
$y^{-n}y' \to \dfrac{v'}{1-n}, \quad y^{1-n} \to v: \qquad \dfrac{v'}{1-\boxed{n}} + \boxed{p}\,v = \boxed{q}$

↓

5. MULTIPLY BY $(1-n)$, READ NEW PIECES
$v' + \underbrace{(1-\boxed{n})\boxed{p}}_{\text{new } p}\,v = \underbrace{(1-\boxed{n})\boxed{q}}_{\text{new } q}$

↓

6. LESSON 3 ON v
The seven lines, with the new p and new q. μ comes from the new p.

↓

7. BACK-SUBSTITUTE
$y = v^{1/(1-\boxed{n})}$

Homogeneous: $y' = F(y/x)$

EXPLANATION

Pieces

Piece	Where to find it
v	$v = \frac{y}{x}$, so $y = vx$
$F(v)$	the right side, rewritten in $\frac{y}{x}$ only, with $\frac{y}{x} \to v$

Get F: divide the top and the bottom by the highest power of x.

Quick example	Work → answer
$y' = \dfrac{x+y}{x}$	$1 + \frac{y}{x}$: $F = 1 + v$
$y' = \dfrac{x^2+y^2}{xy}$	$\div x^2$: $F = \frac{1+v^2}{v}$
$y' = \dfrac{y}{x} + \dfrac{x}{y}$	$F = v + \frac{1}{v}$

Trap: y' is not v'. $y' = v + xv'$.
Trap: Here "homogeneous" means $F(y/x)$, not a zero right side.

WORK

1. REWRITE
$y' = \boxed{F}\left(\frac{y}{x}\right)$

↓

2. SUBSTITUTE
$y = \boxed{v}\,x \qquad y' = \boxed{v} + x\boxed{v}'$ *(product rule)*

↓

3. PLACE AND CANCEL
$\boxed{v} + xv' = \boxed{F(v)} \;\Rightarrow\; xv' = \boxed{F(v)} - \boxed{v}$

↓

4. SEPARATE (LESSON 4)
$\dfrac{dv}{\boxed{F(v)} - \boxed{v}} = \dfrac{dx}{x}$ $\quad (g(x) = \frac{1}{x},\ h(v) = F(v) - v)$

↓

5. INTEGRATE
Both sides. One $+C$. $\int \frac{dx}{x} = \ln|x|$.

↓

6. BACK-SUBSTITUTE
Replace v with $\frac{y}{x}$. Free y if you can.